R Practice Project 3 – Visual Data Analysis on Books Published

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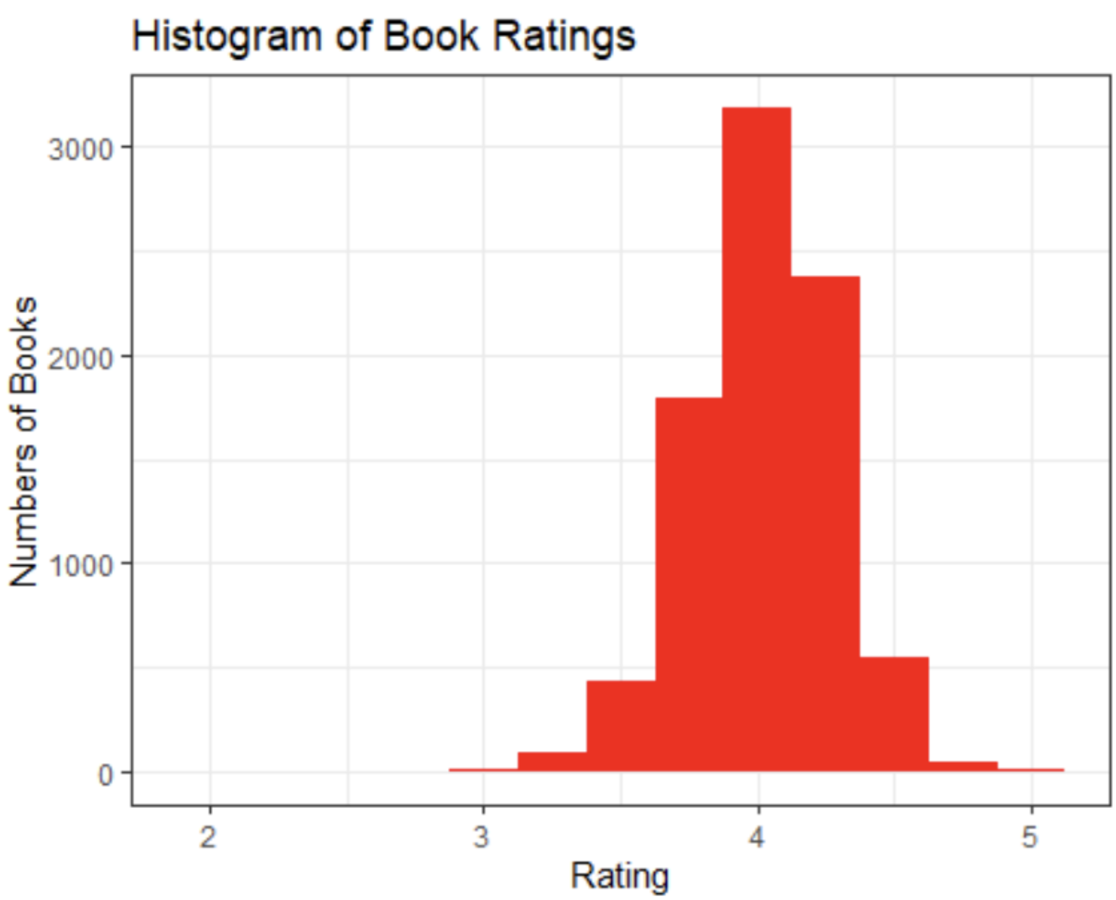
The purpose of ALY6000 Module 3 Project was to examine statistics from Good Reads and Kaggle on authors who have written books throughout history and the many different attributes related towards being an author. For example, categories included awards won, the number of pages, the year it was published, and genre of the book written. More specifically, the project was aimed providing visual analysis of thousands of different data points (books written) and filtering out the paperback copies that were published between 1990 – 2020 and under a page limit of 700 pages. A major key finding from this data presentation was that rate of books tends to have a mean and median of around 4.0 out of 5.0 – furthermore, the average number of pages in a book tends to sit right around 250 – 400 pages with the occasional outlier. Lastly, the total number of books rated per year had a major decrease after 2010 decade started where it peaked at around 600 per year in the start of 2010 and had gone all the way down to under 50 per year by the year 2020.

The initial purpose of the project was to download data on books written, and clean up, as well as filter out any books that did not meet the criteria of what was intended to be analyzed. This included using the lubridate package to convert dates into a standardized month-day-year format (mdy) function to help identify each books date it was first published. As previously mentioned, the data was filtered out to only include books that were written between 1990 and 2020, as well as books that are fewer than 700 pages. Additionally, any rows that contained a NA (not available) data point were removed and columns that were not necessary for the analysis of the data were also taken out – these included the publish_date, edition, characters, price, genres, setting, and isbn. The main point of cleaning up the dataset before looking at it is to make it visually appealing and also easy to comprehend for any audience that is trying to understand the study and statistical significance of looking at this data. Looking at bar graphs, box plots, and other charts with understandable and easy-to-interpret data makes the study more likely to have a practical application and impact on the real world.

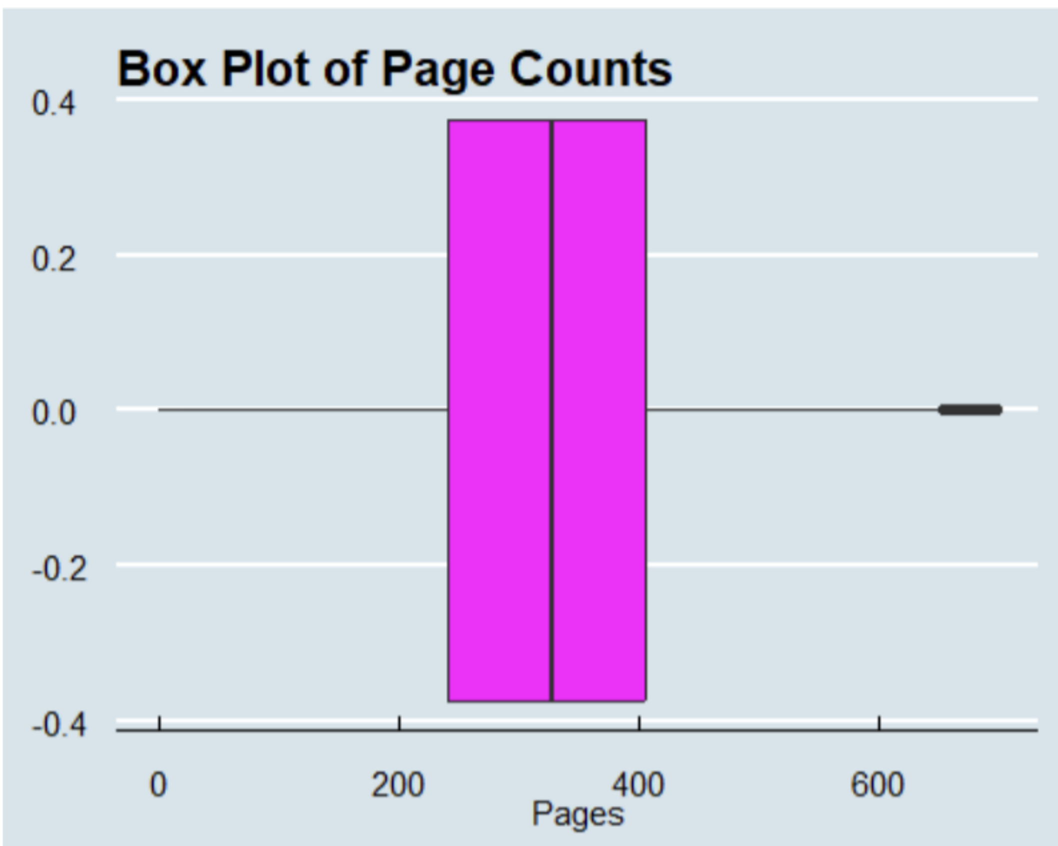
The data analysis section of project 3 is where the importance of visual analysis for making educated conclusions based on the information in a dataset is relevant. For example, a histogram was created based on the number of books and what their rating was.



The average book rating being a 4.0 out of 5.0 was one of the main summarizing points that was taken away from the data analysis of this project. As seen in the histogram, the y-axis scale of number of books goes all the way up to 3000 and the three tallest categories are all either 4.0

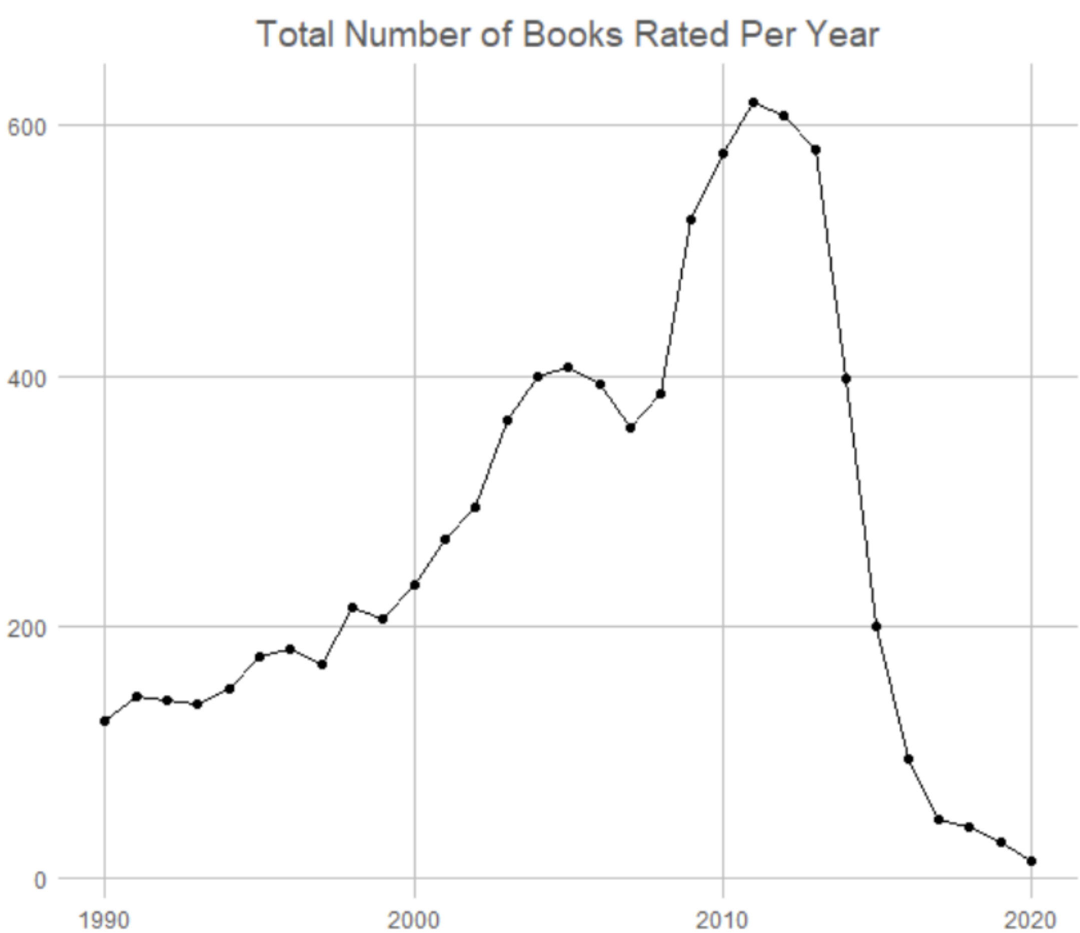
or within a 0.5 range above or below that rating (meaning that it is very close to a 4.0 rating). The histogram provides visually appealing data because it is easy to read and also allows the audience to make the quick conclusion that, though there may be a few outliers, their abundance is so small compared to the number of books rated around 4.0 – therefore, it was be a reasonable conclusion that an average or expected book rating is right around 4.0 or somewhat close to that value.

Furthermore, a box plot of page counts was created to visually analyze the number of pages that a typical book in this dataset had. This was another one of the main takeaways from the project which was that most books have around 250 – 400 pages.



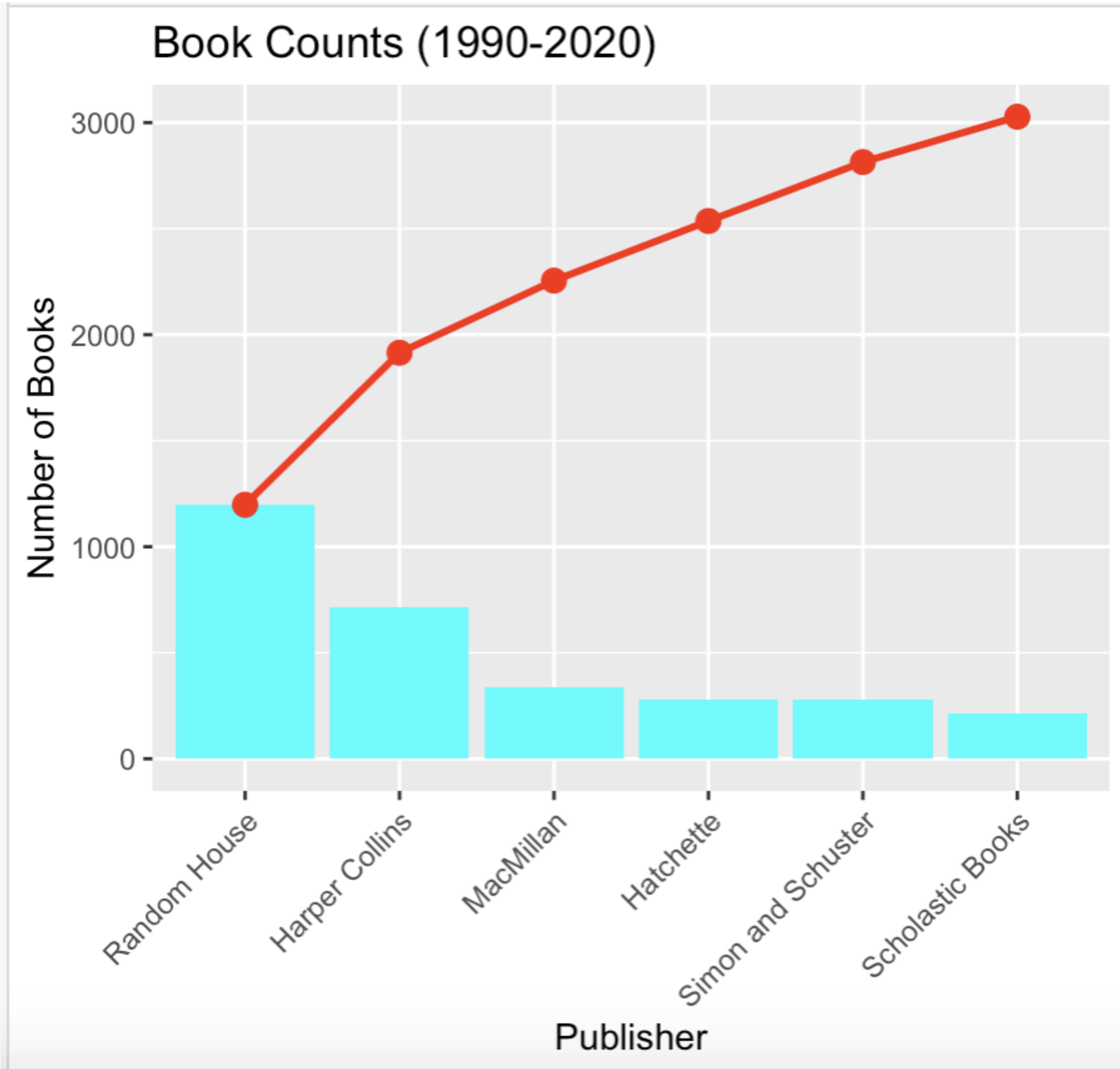
Based on the data being visually presented, it is very clear that majority of books have that given range of 250 – 400 pages due to the purple data being shown in that number range with a few outliers. This generally means that authors like to keep their books not too short but also not too long, likely because the audience wants that desired range due to the inability to stay engaged or an author's inability to create a plot that can go on for longer than 400 pages.

The number of books rated per year was one of the bigger takeaways from this project because the number went from around 600 at the start of 2010 to under 50 by the year 2020 came around. This data is demonstrated visually and clearly on a chart showing the pattern over the years of total number of books rated.



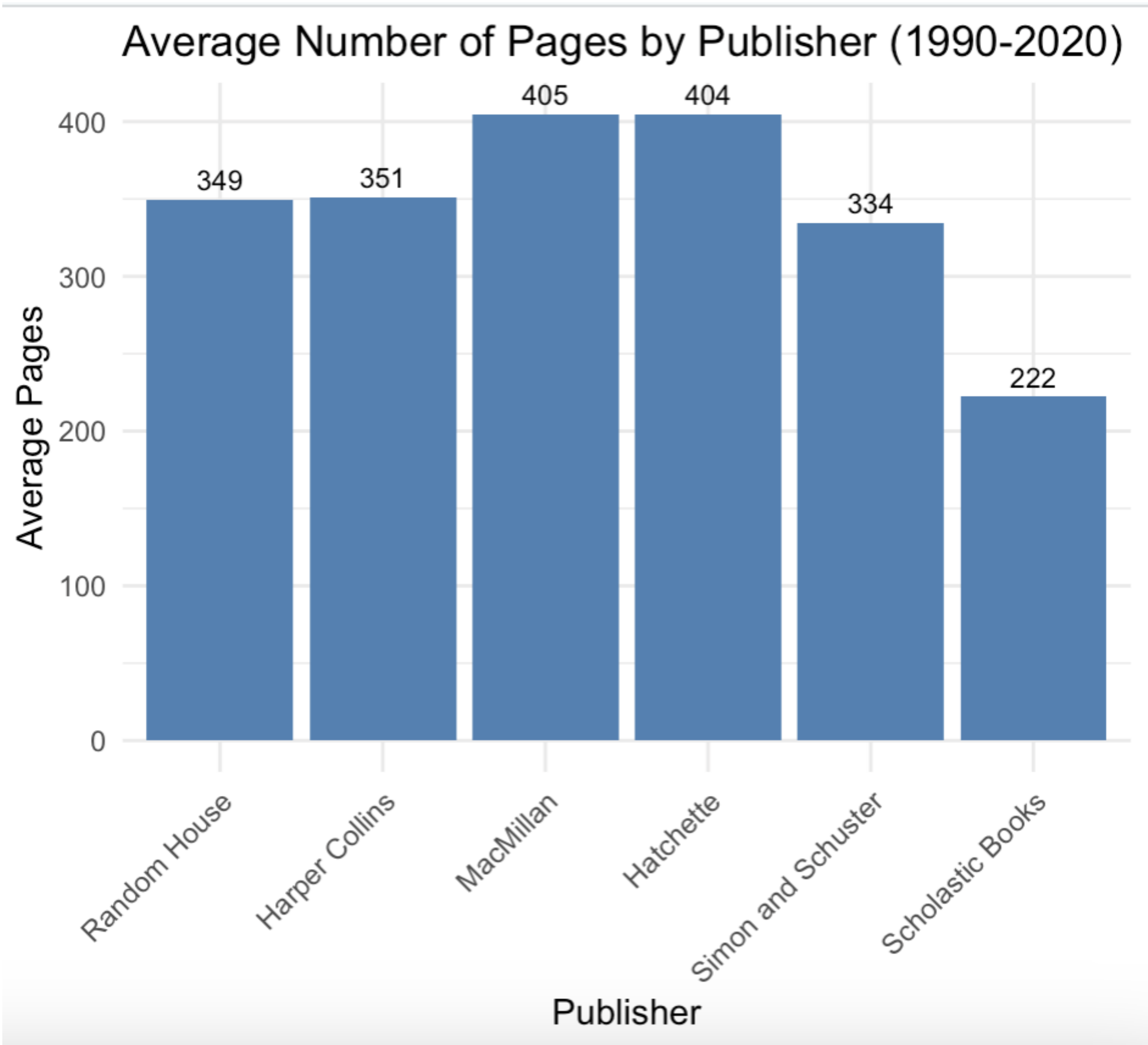
This graph shows the importance of being presented with visual information because it is so clear that, by the shape of the graph, there is a sudden increase and then decrease similar to a climax that occurred right around 2010. The explanation for this is unclear but could be due to the number of reasons including less complete data because it is more recent, an increase in technology usage leading to less paperback books being published, and also the platform providing the data such as Kaggle or Goodreads not being completely up to date. Regardless of the reason, the audience is able to understand that from 1990 to 2010 the number of books rated per year increased somewhat gradually, up until 2010 when it skyrocketed downward in a few short years.

The project then moved on to the analysis of book publishers with the number of books published being over 125, which narrowed it down to only 6 publishers. These were organized in descending order and included Random House, Harper Collins, MacMillan, Hatchette, Simon and Schuster, and Scholastic Books. In order to further analyze the data from these 6 publishers, a column for the cumulative sum and relative frequency were added to the book count column – furthermore, a cumulative sum was added to the relative frequency column. After cleaning up the data, it was presented in a visually appealing manner with a Pareto Chart with an ogive of cumulative Publisher Book Counts. This data was put into a visually appealing form which showed that of the 6 publishers that met our criteria, Random House met the criteria for 1198 of their 1198 books published in this time period. On the other hand, Scholastic Books only met the criteria for 214 of their 3028 books published overall – the trend seemed to be with the larger publishers such as Simon and Schuster and Scholastic Books, a smaller percentage of their books published were rated in our data analysis. The chart shows the differences between number of books rated in our study and the overall book counts for that publisher.



The project also focused on showing visual data in a bar chart form to demonstrate the differences in page amounts for all of these 6 publishers that were part of the second set of analysis questions. For example, the large publishing companies such as MacMillan and Hatchette had an average number of pages right around 405 per book from the years 1990 –

2020. This is drastically different from Scholastic Books which is sitting right around 220, which is almost 200 pages less than these other two companies.



On paper, the numbers 220 and 405 may not seem like that drastic of a difference – however, based on the visual analysis from the bar chart, it is clear that Hatchette and Simon and Schuster wrote books that were significantly longer than Scholastic Books from 1990 – 2020. The main findings from all of this data is that good comprehensible visual analysis can show the major differences between categories being analyzed in datasets. Many of these conclusions generated from the graphs or charts in books study are ones that make sense just based off of looking at the information being presented. Regarding the takeaways from the actual data being presented – there was thousands of books that were written from 1990 – 2020 and many of them were immediately filtered out by the year or page limit criteria. Despite the criteria only needing 125 books published in that 20-year time frame, there were only 6 prominent publishers for paperback books. Ultimately, this was a great visual analysis of different attributes related to books being published over the last 20 years.

References

Kabacoff, R. I (2022). *R in Action: Data analysis and graphics with R and tidyverse (3rd ed.)*. Manning Publication. ISBN 978-1-617-29605-5